

| Calculations                                                         |           |           |           |           |
|----------------------------------------------------------------------|-----------|-----------|-----------|-----------|
|                                                                      | Sheep     | Pig       | Cow       | Goat      |
| Meat Diet Portion                                                    | 1.12%     | 3.61%     | 3.43%     | 0.94%     |
| Meat MCal Target                                                     | 10,180.40 | 32,899.46 | 31,338.35 | 8,619.29  |
| Potential Dairy Value attached to each juvenile                      | 132.54    | 0.00      | 1015.20   | 126.90    |
| Value per juvenile                                                   | 48.00     | 160.00    | 640.00    | 48.00     |
| Adjusted Dairy Value attached to juvenile based on Dairy Utilization | 132.54    | 0.00      | 1015.20   | 126.90    |
| Culled Mothers Value                                                 | 14.77     | 4.41      | 105.26    | 14.77     |
| Culled Adult Males                                                   | 0.74      | 0.22      | 5.26      | 0.74      |
| Final Value per juvenile                                             | 63.51     | 164.63    | 750.53    | 63.51     |
| Value MCals                                                          | 63.51     | 164.63    | 750.53    | 63.51     |
| Offspring required Year <sup>-1</sup>                                | 160.30    | 199.84    | 41.76     | 135.72    |
| MCals from meat                                                      | 10180.40  | 32899.46  | 31338.35  | 8619.29   |
| Max Utilized Mcals allowed from Dairy                                | 21246.40  | 0.00      | 42389.85  | 17222.92  |
| MCals actually Utilized from Dairy                                   | 21246.40  | 0.00      | 42389.85  | 17222.92  |
| Percent of diet from their meat                                      | 1.12%     | 3.61%     | 3.43%     | 0.94%     |
| Percent of Diet from their Dairy                                     | 2.33%     | 0.00%     | 4.65%     | 1.89%     |
| Kg Fleece                                                            | 1,212.77  | 0.00      | 0.00      | 0.00      |
| 1 Year+ (age) skins                                                  | 59.86     | 7.90      | 8.20      | 50.73     |
| Yearly Offspring skins                                               | 292.23    | 236.03    | 61.87     | 247.51    |
| Birthing Events per year                                             | 1.00      | 1.86      | 1.00      | 1.00      |
| Offspring Mother <sup>-1</sup> Year <sup>-1</sup>                    | 0.56      | 5.48      | 0.68      | 0.56      |
| Mothers Step 1                                                       | 286.25    | 36.50     | 61.40     | 242.36    |
| Males Step 1                                                         | 80.15     | 99.92     | 20.88     | 67.86     |
| Females Step 1                                                       | 80.15     | 99.92     | 20.88     | 67.86     |
| Replacement Mothers per year                                         | 57.55     | 7.60      | 7.86      | 48.77     |
| Breeding Males Required                                              | 15.03     | 1.92      | 3.22      | 12.72     |
| Additional Mothers                                                   | 235.59    | 6.61      | 29.57     | 199.63    |
| Males Step 2                                                         | 65.97     | 18.09     | 10.06     | 55.90     |
| Females 2                                                            | 65.97     | 18.09     | 10.06     | 55.90     |
| Total Mothers                                                        | 343.80    | 44.10     | 69.27     | 291.13    |
| Total Males                                                          | 146.12    | 118.01    | 30.93     | 123.76    |
| Total Females                                                        | 14.18     | 81.83     | 10.82     | 11.96     |
| Total Offspring                                                      | 292.23    | 236.03    | 61.87     | 247.51    |
| Total Animals                                                        | 519.13    | 245.86    | 114.25    | 439.57    |
| Units per Hectare                                                    | 1.00      |           |           |           |
| Total Adults per Hectare                                             | 2.18      | 0.33      | 0.53      | 2.18      |
| Total Offspring per Hectare                                          | 1.07      | 1.46      | 0.32      | 1.07      |
|                                                                      | 0.77%     |           | 2.17%     | 0.65%     |
| Total Adults                                                         | 594       | 53        | 102       | 503       |
| Total Offspring                                                      | 292       | 236       | 62        | 248       |
| Days for Maintenance                                                 | 5.00      | 220.00    | 0.00      | 5.00      |
| Dry Mothers                                                          | 51.57     | 0.00      | 7.40      | 43.62     |
| Other                                                                | 4,675.76  | 0.00      | 0.00      | 3,960.23  |
| MCals Dry Mothers                                                    | 60,232.69 | 0.00      | 15,670.28 | 50,942.62 |
| total Maintenance MCals                                              | 64,908.45 | 0.00      | 15,670.28 | 54,902.85 |

| MCals Required                  |           |           |           |           |
|---------------------------------|-----------|-----------|-----------|-----------|
|                                 | Sheep     | Pig       | Cow       | Goat      |
| Gestating                       | 174,026   | 107,758   | 103,563   | 147,395   |
| Lactating                       | 332,008   | 38,307    | 111,358   | 281,201   |
| Maintenance                     | 64,908    | 0         | 15,670    | 54,903    |
| Adult Males                     | 17,553    | 0         | 6,825     | 14,861    |
| Offspring                       | 80,218    | 237,207   | 226,428   | 67,943    |
| TOTAL                           | 668,713.7 | 383,272.7 | 463,844.3 | 566,302.6 |
| Animal Diet Breakdown           |           |           |           |           |
|                                 | Sheep     | Pig       | Cow       | Goat      |
| Crop                            | Barley    | Barley    | Barley    | Barley    |
| Grams <sup>-1</sup> Straw/Chaff | 1000.0    | 0.0       | 4956.0    | 1000.0    |
| MCal kg <sup>-1</sup>           |           |           |           |           |
| Grams <sup>-1</sup> Grain       | 750.0     | 750.0     | 1500.0    | 750.0     |
| MCal kg <sup>-1</sup>           |           |           |           |           |
| Days                            | 90.0      | 90.0      | 90.0      | 90.0      |
| Number of Animals               | 292.2     | 236.0     | 61.9      | 247.5     |
| MCals from Straw/Chaff          | 37347.6   | 0.0       | 39184.3   | 31632.3   |
| MCals from Grain                | 59177.5   | 47795.5   | 25055.6   | 50121.6   |
| MCals from other feed           | 96525.1   | 47795.5   | 64239.8   | 81754.0   |

| Diet Calculations                           |                  |         |
|---------------------------------------------|------------------|---------|
| Plants Mcal                                 | 740,390.828      |         |
| Meat Mcal                                   | 91,250           |         |
| Dairy                                       | 80,859.172431353 |         |
|                                             | Perfect          |         |
| Plants %                                    | 81.14%           |         |
| Meat %                                      | 10.00%           |         |
| Dairy %                                     | 8.86%            |         |
| check                                       | Perfect          |         |
| Percent                                     |                  |         |
| Domestic Meat                               | 9.10%            |         |
| Dairy                                       | 8.86%            |         |
| Wild Meat                                   | 0.90%            |         |
| Crops                                       | 73.02%           |         |
| Wild Plants                                 | 8.11%            |         |
| check                                       | Perfect          |         |
| new dairy                                   | 80859.17         |         |
| potential dairy < limit                     | TRUE             |         |
| new limit                                   | 90.0%            |         |
| the limit greater than that left to satisfy | TRUE             |         |
| limit                                       | 1                |         |
| limit dairy checked                         | FALSE            |         |
| Mcalories                                   |                  | Surplus |
| Domestic Meat                               | 83,037.500       |         |
| Dairy                                       | 80,859.172       |         |
| Wild Meat                                   | 8212.500         |         |
| Crops                                       | 666,351.745      |         |
| Wild Plants                                 | 74,039.083       |         |
| check                                       | Perfect          |         |
| SURPLUS DAIRY                               | 0.000            |         |
| This could account for additional % of diet | 0.00%            |         |

Herd Modelling

|                                     | Sheep  | Pig    | Cow    | Goat   |  |  |  |  |
|-------------------------------------|--------|--------|--------|--------|--|--|--|--|
| Mothers Req'd                       | 344    | 44     | 69     | 291    |  |  |  |  |
| Males for Slaughter                 | 146    | 118    | 31     | 124    |  |  |  |  |
| Females for Slaughter               | 14     | 82     | 11     | 12     |  |  |  |  |
| Offspring available for Consumption | 292    | 236    | 62     | 248    |  |  |  |  |
| Gestation Req'd                     | 95460  | 91238  | 69898  | 80822  |  |  |  |  |
| Lactation Req'd                     | 182119 | 32435  | 75159  | 154192 |  |  |  |  |
| Maintenance Req'd                   | 216895 | 0      | 20948  | 183689 |  |  |  |  |
| Adult Males                         | 17553  | 0      | 6825   | 14861  |  |  |  |  |
| Offspring Required                  | 95999  | 258450 | 270971 | 81308  |  |  |  |  |
| Energy Required                     | 608026 | 382123 | 443802 | 514873 |  |  |  |  |
| Ewe Fleece Kg Yr-1                  | 774    |        |        |        |  |  |  |  |
| Ram Fleece Kg year-1                | 30     |        |        |        |  |  |  |  |
| Juvenile Fleece Kg Yr-1             | 409    |        |        |        |  |  |  |  |
| Kg Fleece Year-1                    | 1213   | 0      | 0      | 0      |  |  |  |  |